cover">

AP CALCULUS AB · EXAMINATION REPOSITORY

Unit 3 Composite, Implicit & Inverse Functions

100-Question Priority-Starred Master Set

- Questions 1–90: Multiple Choice (Full Priority Ratings)
- Questions 91–100: Free Response with Official AP Scoring Rubrics
- Complete Answer Key and Detailed Worked Solutions at the End

OFFICIAL AP CED ALIGNED · MIXED LEVEL · RIGOROUSLY VERIFIED

Examination Structure

This paper contains 100 continuously numbered questions. All answers, worked solutions, and scoring rubrics appear at the end of the document.

⭐ Priority Rating System

★★★★★	**Very Important**	High-frequency exam concepts: Chain Rule with composites, implicit tangent lines, inverse derivative rule, inverse trig with chain.
★★★	**Mandatory**	Essential skills: standard chain rule applications, higher-order derivatives, selecting the right procedure.
★★	**Important**	Foundational practice: single-step chain rule and memorised inverse trig derivative formulas.

- **Questions 1–90:** Multiple choice, four options, exactly one correct answer.
- **Questions 91–100:** Free response, 9 points each, with analytical setups and rubrics.
- **Difficulty Mix:** Easy 5 · Moderate 68 · Hard 17.
- **Calculator Policy:** No calculator required.
- **Answer Section:** Begins after Question 100 — answer key, then full MCQ solutions, then FRQ solutions with rubrics.

Topic Coverage (Unit 3.1 – 3.6)

3.1 The Chain Rule

3.2 Implicit Differentiation

3.3 Differentiating Inverse Functions

3.4 Inverse Trigonometric Derivatives

3.5 Selecting Procedures for Derivatives

3.6 Higher-Order Derivatives

> **Study Tip:** The Chain Rule is the backbone of this unit. If a question involves a function inside another function, always ask: what is the inside, and what is its derivative?

ITEM 01

EASY · SINGLE BEST CHOICE

★★ IMPORTANT

If $f(x) = (3x + 1)^4$, then $f'(x) =$

A $4(3x + 1)^3$

B $3(3x + 1)^3$

C $12(3x + 1)^3$

D $12(3x + 1)^4$

ITEM 02

EASY · SINGLE BEST CHOICE

★★ IMPORTANT

$d/dx\ [\ \sin(5x)\ ] =$

A $5\sin(5x)$

B $\cos(5x)$

C $-5\cos(5x)$

D $5\cos(5x)$

ITEM 03

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\ [\ \cos(x^2)\ ] =$

A $-\sin(x^2)$

B $2x\sin(x^2)$

C $-2x\sin(x^2)$

D $-2x\cos(x^2)$

ITEM 04

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If f(x) = e^(3x²), then f′(x) =

A 3x·e^(3x²)

B 6x·e^(3x)

C 6x·e^(3x²)

D e^(3x²)

ITEM 05

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

d/dx [ln(5x² + 1)] =

A 10x/(5x² + 1)

B 5x/(5x² + 1)

C 1/(5x² + 1)

D 10x

ITEM 06

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If f(x) = √(x³ + 1), then f′(1) =

A 3/4

B 3√2 / 2

C 3/2

D 3√2 / 4

ITEM 07

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

$d/dx\ [\ \tan(3x)\ ] =$

A $3\sec^2(3x)$

B $3\sec(3x)\tan(3x)$

C $3\tan^2(3x)$

D $\sec^2(3x)$

ITEM 08

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = (x^2 - 4)^5$, then $f'(2) =$

A 20

B 80

C 160

D 0

ITEM 09

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

$d/dx\ [\ \sin^2 x\ ] =$

A $2\sin x \cos x$

B $2\cos x$

C $2\sin x$

D $-2\sin x \cos x$

ITEM 10

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If f(x) = e^(sin x), then f′(π/2) =

A 1

B −e

C e

D 0

ITEM 11

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If f(x) = ln(cos x), then f′(π/4) =

A −1

B 1

C −√2/2

D 0

ITEM 12

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

d/dx [sec(2x)] =

A sec(2x) tan(2x)

B $2\sec^2(2x)$

C 2 sec(2x) tan(2x)

D sec(2x) tan(x)

ITEM 13

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\ [\ \sqrt{(\sin x)}\ ] =$

A $-\cos x / (2\sqrt{(\sin x)})$

B $\cos x / (2\sqrt{(\sin x)})$

C $1 / (2\sqrt{(\sin x)})$

D $\cos x / \sqrt{(\sin x)}$

ITEM 14

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

Let $g(x) = f(x^2)$. If $f'(4) = 3$, then $g'(2) =$

A 12

B 9

C 3

D 6

ITEM 15

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

Let $h(x) = f(g(x))$ with $g(1) = 3$, $g'(1) = 5$, and $f'(3) = 2$. Then $h'(1) =$

A 7

B 10

C 8

D 15

ITEM 16

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\ [\ 1/(3x + 2)\ ] =$

A $-3/(3x + 2)^2$

B $3/(3x + 2)^2$

C $-1/(3(3x + 2)^2)$

D $-1/(3x + 2)^2$

ITEM 17

HARD · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

$d/dx\ [\ \cos^3(2x)\ ] =$

A $3\cos^2(2x)\sin(2x)$

B $-6\cos^2(2x)\sin(2x)$

C $-3\cos^2(2x)\sin(2x)$

D $6\cos^2(2x)\sin(2x)$

ITEM 18

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\ [\ e^{\wedge}(\cos x)\ ] =$

A $-\sin x \cdot e^{\wedge}(\cos x)$

B $-e^{\wedge}(\sin x)$

C $\sin x \cdot e^{\wedge}(\cos x)$

D $e^{\wedge}(\cos x)$

ITEM 19

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = \ln(x^2 + 1)$, then $f'(1) =$

A 1/2

B 0

C 2

D 1

ITEM 20

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\ [\ \sin(\ln x)\ ] =$

A cos(ln x) / x

B 1 / (x cos(ln x))

C cos(ln x)

D −sin(ln x) / x

ITEM 21

HARD · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $f(x) = (x + 1/x)^3$, then $f'(1) =$

A 24

B 12

C 0

D 3

ITEM 22

HARD · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\ [\ 2^{(3x)}\ ] =$

A $3 \cdot 2^{(3x)}$

B $3 \ln 2 \cdot 2^{(3x)}$

C $2^{(3x)} \ln 2$

D $3x \cdot 2^{(3x-1)}$

ITEM 23

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\ [\ \tan(x^2)\ ] =$

A $\sec^2(2x)$

B $\sec^2(x^2)$

C $2x \tan^2(x^2)$

D $2x \sec^2(x^2)$

ITEM 24

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $f(x) = \sqrt{(4 - x^2)}$, then $f'(1) =$

A $-\sqrt{3}/3$

B $\sqrt{3}/3$

C $-2\sqrt{3}/3$

D $-1/3$

ITEM 25

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\ [\ \ln(x^3 + 2x)\ ] =$

A $(3x^2 + 2)/(x^3 + 2x)$

B $3x^2/(x^3 + 2x)$

C $1/(x^3 + 2x)$

D $3x^2 + 2$

ITEM 26

EASY · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $x^2 + y^2 = 25$, then $dy/dx =$

A y/x

B $-y/x$

C $-x/y$

D x/y

ITEM 27

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

For the circle $x^2 + y^2 = 25$, the slope of the tangent line at the point $(3, 4)$ is

A $-3/4$

B $-4/3$

C $3/4$

D $4/3$

ITEM 28

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $xy = 6$, then $dy/dx =$

A $-y/x$

B $-x/y$

C x/y

D y/x

ITEM 29

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $x^2 + xy = 4$, then $dy/dx =$

A $-2x/y$

B $(2x + y)/x$

C $-(2x + y)/x$

D $-y/(2x)$

ITEM 30

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If $y^2 = x^3 + 1$, then $dy/dx =$

A $-3x^2/(2y)$

B $3x^2/y$

C $3x^2/(2y)$

D $2y/(3x^2)$

ITEM 31

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If $\sin y = x$, then $dy/dx =$

A 1/sin y

B 1/cos y

C cos y

D −1/cos y

ITEM 32

HARD · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $e^y = x^2$ for $x > 0$, then $dy/dx =$

A x/2

B 2/x

C 2x

D 2x·e^y

ITEM 33

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $x^3 + y^3 = 9$, then dy/dx at the point $(1, 2)$ is

A −1/4

B −1/2

C −4

D 1/4

ITEM 34

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If $x^2y = 4$, then dy/dx at the point $(2, 1)$ is

A -1

B -2

C $-1/2$

D 1

ITEM 35

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

On the circle $x^2 + y^2 = 25$, the tangent line is horizontal at

A $(3, 4)$ only

B $(0, 5)$ and $(0, -5)$

C $(5, 0)$ and $(-5, 0)$

D the origin only

ITEM 36

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

On the circle $x^2 + y^2 = 25$, the tangent line is vertical at

A nowhere on the circle

B $(5, 0)$ and $(-5, 0)$

C $(3, 4)$ only

D $(0, 5)$ and $(0, -5)$

ITEM 37

HARD · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $x^2 + y^2 = 1$, then $d^2y/dx^2 =$

A $-1/y^2$

B $1/y^3$

C $-x/y$

D $-1/y^3$

ITEM 38

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

For the ellipse $4x^2 + 9y^2 = 36$, the slope of the tangent line at $(0, 2)$ is

A $2/3$

B 0

C $-4/9$

D undefined

ITEM 39

HARD · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $\ln y + x = y$, then $dy/dx =$

A $y/(y - 1)$

B $1/(y - 1)$

C $y/(1 - y)$

D $(y - 1)/y$

ITEM 40

HARD · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $y = \sin(xy)$, then $dy/dx =$

A $\cos(xy)$

B $y\cos(xy) / (1 - x\cos(xy))$

C $x\cos(xy) / (1 - y)$

D $y\cos(xy)$

ITEM 41

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If $xy + y^2 = 2$, then $dy/dx =$

A $y/(x + 2y)$

B $-y/(x + y)$

C $-y/(x + 2y)$

D $-x/(x + 2y)$

ITEM 42

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If $x = y^3 + y$, then $dy/dx =$

A $1/(3y^2 + 1)$

B $1/(3y^2)$

C $-1/(3y^2 + 1)$

D $3y^2 + 1$

ITEM 43

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

An equation of the line tangent to $x^2 + y^2 = 25$ at $(3, 4)$ is

A $y - 4 = -(3/4)(x - 3)$

B $y = -(3/4)x$

C $y - 4 = (3/4)(x - 3)$

D $y - 3 = -(3/4)(x - 4)$

ITEM 44

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If $\cos y = x$, then $dy/dx =$

A $-1/\sin y$

B $-\sin y$

C $1/\cos y$

D $1/\sin y$

ITEM 45

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If $x^2 - y^2 = 16$, then dy/dx at the point $(5, 3)$ is

A $5/3$

B $-3/5$

C $-5/3$

D $3/5$

ITEM 46

HARD · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If x·e^y = 1, then dy/dx =

A −1/(x e^y)

B −e^y

C 1/x

D −1/x

ITEM 47

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If f is one-to-one and differentiable, then $(f^{-1})'(b)$ =

A $1/f'(f^{-1}(b))$

B $-1/f'(b)$

C $f'(f^{-1}(b))$

D $1/f'(b)$

ITEM 48

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

Let $f(x) = x^3 + x + 1$ and let $g = f^{-1}$. Then $g'(3)$ =

A 3

B 1/4

C 4

D 1/28

ITEM 49

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

Let $f(x) = x^5 + 2x - 1$. Then $(f^{-1})'(2) =$

A 1/7

B 1/3

C 1/2

D 7

ITEM 50

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

A differentiable function f satisfies $f(2) = 5$ and $f'(2) = 3$. Then $(f^{-1})'(5) =$

A 5

B 3

C 1/5

D 1/3

ITEM 51

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

Let $f(x) = e^x$ and let $g = f^{-1}$. Then $g'(e) =$

A $1/e^2$

B 1

C e

D 1/e

ITEM 52

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

Let $f(x) = \sqrt{x}$ for $x \geq 0$ and let $g = f^{-1}$. Then $g'(2) =$

A $1/4$

B $1/(2\sqrt{2})$

C 4

D 2

ITEM 53

HARD · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

Let $f(x) = 2x + \sin x$. Then $(f^{-1})'(0) =$

A 1

B $1/3$

C 3

D $1/2$

ITEM 54

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If f is invertible with $f(3) = 7$ and $f'(3) = 1/2$, then $(f^{-1})'(7) =$

A $1/7$

B 2

C 7

D $1/2$

ITEM 55

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The table gives values for a differentiable, increasing function f.

x	f(x)	f′(x)
1	4	2
2	7	5
3	10	8

If $g = f^{-1}$, then $g'(7) =$

A 5

B 1/2

C 1/5

D 1/8

ITEM 56

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

Let $f(x) = x^3 + 2$. Then $(f^{-1})'(10) =$

A 12

B 1/3

C 1/12

D 1/10

ITEM 57

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

Suppose f is differentiable with $f'(x) > 0$ for all x, and $g = f^{-1}$. Then $g'(x)$ is

A negative for all x in the domain of g

B positive for all x in the domain of g

C always undefined

D always equal to zero

ITEM 58

HARD · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

Let $f(x) = \tan x$ on $(-\pi/2, \pi/2)$ and let $g = f^{-1}$. Then $g'(1) =$

A 2

B 1/2

C $\sqrt{2}/2$

D 1

ITEM 59

EASY · SINGLE BEST CHOICE

★★ IMPORTANT

$d/dx\,[\arcsin x] =$

A $1/\sqrt{(1 - x^2)}$

B $-1/\sqrt{(1 - x^2)}$

C $1/(1 + x^2)$

D $-1/(1 + x^2)$

ITEM 60

EASY · SINGLE BEST CHOICE

★★ IMPORTANT

$d/dx\,[\arctan x] =$

A $1/(1 - x^2)$

B $-1/(1 + x^2)$

C $1/(1 + x^2)$

D $1/\sqrt{(1 - x^2)}$

ITEM 61

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\ [\ \arccos x\ ] =$

A $1/\sqrt{(x^2 - 1)}$

B $1/\sqrt{(1 - x^2)}$

C $-1/(1 + x^2)$

D $-1/\sqrt{(1 - x^2)}$

ITEM 62

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

$d/dx\ [\ \arctan(2x)\ ] =$

A $2/(1 + 4x^2)$

B $2/(1 + 2x^2)$

C $1/(1 + 4x^2)$

D $2/(1 + x^2)$

ITEM 63

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

$d/dx\ [\ \arcsin(x^2)\ ] =$

A $2x/\sqrt{(1 - x^4)}$

B $2x/\sqrt{(1 - x^2)}$

C $2x/(1 + x^4)$

D $1/\sqrt{(1 - x^4)}$

ITEM 64

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = \arctan x$, then $f'(1) =$

A 1

B 2

C $\pi/4$

D 1/2

ITEM 65

HARD · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

$d/dx\ [\ \arcsin(x/3)\ ] =$

A $1/\sqrt{(1 - 9x^2)}$

B $1/(3\sqrt{(1 - x^2)})$

C $3/\sqrt{(9 - x^2)}$

D $1/\sqrt{(9 - x^2)}$

ITEM 66

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

$d/dx\ [\ x \cdot \arctan x\ ] =$

A $\arctan x + x/(1 + x^2)$

B $x/(1 + x^2)$

C $\arctan x + 1/(1 + x^2)$

D $1/(1 + x^2)$

ITEM 67

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\ [\ \arccos(2x)\ ] =$

A $2/\sqrt{(1 - 4x^2)}$

B $-2/\sqrt{(1 - 2x^2)}$

C $-2/\sqrt{(1 - 4x^2)}$

D $-1/\sqrt{(1 - 4x^2)}$

ITEM 68

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\ [\ e^{(\arctan x)}\ ] =$

A $e^{(\arctan x)/(1 + x^2)}$

B $e^{(\arctan x)}$

C $e^{(1/(1 + x^2))}$

D $\arctan x \cdot e^{(\arctan x)}$

ITEM 69

HARD · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

$d/dx\ [\ \arctan(1/x)\ ] =$

A $1/(1 + x^2)$

B $-1/x^2$

C $-1/(1 + x^2)$

D $-1/(x^2(1 + x^2))$

ITEM 70

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\ [\ \arcsin(e^x)\ ] =$

A $e^x/\sqrt{(1 - e^x)}$

B $e^x/\sqrt{(1 - e^{(2x)})}$

C $1/\sqrt{(1 - e^{(2x)})}$

D $e^x/(1 + e^{(2x)})$

ITEM 71

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\ [\ \arctan(x^2)\ ] =$

A $2x/(1 + x^2)$

B $2x/(1 + x^4)$

C $1/(1 + x^4)$

D $2x/\sqrt{(1 - x^4)}$

ITEM 72

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = \arcsin x$, then $f'(1/2) =$

A $1/2$

B $\sqrt{3}/3$

C $2\sqrt{3}/3$

D $4/3$

ITEM 73

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = \arctan(3x)$, then $f'(0) =$

A 3

B 0

C 1/3

D 1

ITEM 74

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = x^4$, then $f'''(x) =$

A $12x^2$

B $4x^3$

C 24

D 24x

ITEM 75

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = \sin x$, then $f''(x) =$

A $\cos x$

B $-\sin x$

C $-\cos x$

D $\sin x$

ITEM 76

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If f(x) = e^(2x), then f″(x) =

A 4e^(2x)

B e^(2x)

C $4e^{x}$

D 2e^(2x)

ITEM 77

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

If f(x) = ln x, then f″(x) =

A $2/x^3$

B $-1/x^2$

C $1/x^2$

D 1/x

ITEM 78

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If f(x) = 1/x, then f″(x) =

A $-2/x^3$

B $1/x^2$

C $-1/x^2$

D $2/x^3$

ITEM 79

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

A particle moves with position $s(t) = t^3 - 6t^2$. Its acceleration at $t = 1$ is

A −9

B 6

C −3

D −6

ITEM 80

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

If $f(x) = x^3 + 2x^2$, then $f''(1) =$

A 7

B 6

C 4

D 10

ITEM 81

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

$\frac{d^2}{dx^2}[\cos(2x)] =$

A $4\cos(2x)$

B $-2\sin(2x)$

C $-4\cos(2x)$

D $-\cos(2x)$

ITEM 82

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

$d/dx\,[\,x^2 \sin(3x)\,] =$

A $6x^2 \cos(3x)$

B $2x \sin(3x) + 3x^2 \cos(3x)$

C $2x \cos(3x)$

D $2x \sin(3x) + x^2 \cos(3x)$

ITEM 83

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\,[\,e^x/(x+1)\,] =$

A $-e^x/(x+1)^2$

B $x \cdot e^x/(x+1)^2$

C $e^x/(x+1)^2$

D $e^x(x+2)/(x+1)^2$

ITEM 84

MODERATE · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\,[\,\sqrt{x} \cdot \ln x\,] =$

A $(\ln x)/(2\sqrt{x}) + 1/\sqrt{x}$

B $1/(2x\sqrt{x})$

C $(\ln x)/(2\sqrt{x})$

D $(\ln x + 1)/(2\sqrt{x})$

ITEM 85

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

The table gives values for differentiable functions f and g.

x	f(x)	f′(x)	g(x)	g′(x)
1	2	3	−1	6
2	1	7	1	5

If $h(x) = f(g(x))$, then $h'(2) =$

A 35

B 15

C 7

D 12

ITEM 86

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

Using the same table as the previous question, if $k(x) = g(f(x))$, then $k'(2) =$

A 21

B 35

C 42

D 15

ITEM 87

HARD · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

$\frac{d}{dx}\left[\ln(x \cdot e^x)\right] =$

A $1 + x$

B $1/x + 1$

C $1/x$

D e^x/x

ITEM 88

HARD · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

$d/dx\ [\ e^{(2 \ln x)}\ ]$ for $x > 0 =$

A $2/x$

B $e^{(2 \ln x)}/x$

C x^2

D $2x$

ITEM 89

HARD · SINGLE BEST CHOICE

★★★ MANDATORY

$d/dx\ [\ (2x + 1)^3(x - 1)\ ] =$

A $6(2x + 1)^2(x - 1)$

B $6(2x + 1)^2(x - 1) + (2x + 1)^3$

C $3(2x + 1)^2(x - 1) + (2x + 1)^3$

D $3(2x + 1)^2$

ITEM 90

MODERATE · SINGLE BEST CHOICE

★★★★★ VERY IMPORTANT

$d/dx\ [\ \sin(\cos x)\ ] =$

A $\sin x \cdot \cos(\cos x)$

B $\cos(\cos x)$

C $-\cos(\sin x)$

D $-\sin x \cdot \cos(\cos x)$

Question 91: Composite Functions from a Table

★★★★★ VERY IMPORTANT

9 Points

The functions f and g are differentiable for all real numbers. Selected values are given in the table below.

x	f(x)	f′(x)	g(x)	g′(x)
1	3	5	2	−4
2	1	7	3	6
3	2	−2	1	8

(a) [3 Points] Let h(x) = f(g(x)). Find h′(1). Show the work that leads to your answer.

(b) [3 Points] Let k(x) = g(f(x)). Find k′(1). Show the work that leads to your answer.

(c) [3 Points] Let m(x) = f(x)·g(x). Find m′(3).

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 92: Implicit Differentiation and Tangent Lines

★★★★★ VERY IMPORTANT | 9 Points

Consider the curve defined by $x^2 + xy + y^2 = 7$. The point $(1, 2)$ lies on this curve.

(a) [3 Points] Show that $dy/dx = -(2x + y)/(x + 2y)$.

(b) [3 Points] Write an equation for the line tangent to the curve at the point $(1, 2)$.

(c) [3 Points] Find the coordinates of all points on the curve at which the tangent line is horizontal.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 93: Derivative of an Inverse Function

★★★★★ VERY IMPORTANT

9 Points

Let f be the function defined by $f(x) = x^3 + 2x - 1$, and let g be the inverse function of f.

(a) [2 Points] Show that f is increasing for all real x, and explain why this guarantees f has an inverse.

(b) [4 Points] Find g(2) and g′(2). Show the reasoning that leads to your answers.

(c) [3 Points] Write an equation for the line tangent to the graph of y = g(x) at x = 2.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 94: Inverse Trigonometric Derivatives

★★★★★ VERY IMPORTANT

9 Points

Consider the functions $h(x) = \arctan(x^2)$, $p(x) = \arcsin(3x)$, and $q(x) = x \cdot \arccos x$.

(a) [3 Points] Find $h'(x)$ and evaluate $h'(1)$.

(b) [3 Points] Find $p'(x)$ and state the open interval of x-values for which p is differentiable.

(c) [3 Points] Find $q'(x)$ and evaluate $q'(0)$.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 95: Higher-Order Derivatives and Particle Motion

★★★★★ VERY IMPORTANT | 9 Points

A particle moves along a horizontal line so that its position at time $t \geq 0$ is $s(t) = t^3 - 6t^2 + 9t$.

(a) [2 Points] Find the velocity $v(t)$ and write it in factored form.

(b) [2 Points] Find the acceleration $a(t)$ and evaluate $a(2)$.

(c) [2 Points] For what values of t is the particle at rest?

(d) [3 Points] At $t = 2.5$, is the speed of the particle increasing or decreasing? Justify your answer.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 96: Selecting Procedures: Product with Chain Rule

★★★★★ VERY IMPORTANT | 9 Points

Let f be the function defined by f(x) = e^(2x)·sin(3x).

(a) [3 Points] Find f′(x).

(b) [2 Points] Evaluate f′(0).

(c) [4 Points] Find f″(x) and evaluate f″(0).

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 97: Implicit Curve with Horizontal and Vertical Tangents

★★★★★ VERY IMPORTANT | 9 Points

Consider the curve defined by $x^2 + y^2 = 4x$.

(a) [3 Points] Find dy/dx in terms of x and y.

(b) [3 Points] Find the coordinates of all points on the curve where the tangent line is horizontal.

(c) [3 Points] Find the coordinates of all points on the curve where the tangent line is vertical.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 98: Applied Chain Rule: Exponential Growth

★★★ MANDATORY

9 Points

The number of bacteria in a culture is modeled by $N(t) = 1200 \cdot e^{(0.3t)}$, where t is measured in hours and $0 \leq t \leq 10$.

(a) [2 Points] Find $N'(t)$.

(b) [3 Points] Find $N'(2)$ and interpret the meaning of this value in the context of the problem, including units.

(c) [4 Points] Find the time t at which the bacteria are growing at a rate of 1080 bacteria per hour.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 99: Inverse Function from a Table

★★★★★ VERY IMPORTANT

9 Points

The function f is differentiable and increasing on the interval [0, 3]. Selected values are given in the table.

x	0	1	2	3
f(x)	1	3	7	13
f′(x)	1	3	5	7

(a) [2 Points] Explain why f has an inverse function g on [0, 3].

(b) [2 Points] Find g(7).

(c) [3 Points] Find g′(7). Show your reasoning.

(d) [2 Points] Write an equation for the line tangent to y = g(x) at x = 7.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

Question 100: Composite Trigonometric Function and Second Derivative

★★★★★ VERY IMPORTANT | 9 Points

Let f be the function defined by $f(x) = \cos(x^2)$ for $0 \le x \le \sqrt{\pi}$.

(a) [2 Points] Find $f'(x)$.

(b) [4 Points] Find $f''(x)$.

(c) [3 Points] Find all values of x in $[0, \sqrt{\pi}]$ at which the graph of f has a horizontal tangent line.

Reminder: Show the setup of every step and state any theorem or rule you use by name.

End of Question Paper

You have reached the end of Questions 1 through 100.

Before you continue: Make sure you have genuinely attempted every question. The pages that follow contain the complete answer key, fully worked multiple choice solutions with analysis of every incorrect option, and model free response solutions with AP-style scoring rubrics.

What Comes Next

- **Answer Key:** Letter answers for Questions 1–90 with priority ratings.
- **Multiple Choice Solutions:** Step-by-step reasoning for Questions 1–90 plus an explanation of why each distractor fails.
- **Free Response Solutions:** Complete model answers for Questions 91–100 with point-by-point rubrics.
- **Appendix:** Essential Unit 3 Rules & Procedures formulas and theorems.

Multiple Choice Answer Key

Letter answers with priority star ratings. Questions 91–100 are free response and are scored using the rubrics that follow.

Q01	Q02	Q03	Q04	Q05	Q06	Q07	Q08	Q09	Q10
C ★★	D ★★	C ★★★	C ★★★	A ★★★	D ★★★	A ★★★★★	D ★★★	A ★★★★★	D ★★★
Q11	**Q12**	**Q13**	**Q14**	**Q15**	**Q16**	**Q17**	**Q18**	**Q19**	**Q20**
A ★★★★★	C ★★★	B ★★★	A ★★★★★	B ★★★★★	A ★★★	B ★★★★★	A ★★★	D ★★★	A ★★★
Q21	**Q22**	**Q23**	**Q24**	**Q25**	**Q26**	**Q27**	**Q28**	**Q29**	**Q30**
C ★★★★★	B ★★★	D ★★★	A ★★★★★	A ★★★	C ★★★★★	A ★★★	A ★★★★★	C ★★★★★	C ★★★
Q31	**Q32**	**Q33**	**Q34**	**Q35**	**Q36**	**Q37**	**Q38**	**Q39**	**Q40**
B ★★★	B ★★★★★	A ★★★★★	A ★★★	B ★★★	B ★★★	D ★★★★★	B ★★★	A ★★★★★	B ★★★★★
Q41	**Q42**	**Q43**	**Q44**	**Q45**	**Q46**	**Q47**	**Q48**	**Q49**	**Q50**
C ★★★	A ★★★★★	A ★★★	A ★★★	A ★★★	D ★★★★★	A ★★★★★	B ★★★★★	A ★★★	D ★★★★★
Q51	**Q52**	**Q53**	**Q54**	**Q55**	**Q56**	**Q57**	**Q58**	**Q59**	**Q60**
D ★★★	C ★★★	B ★★★★★	B ★★★	C ★★★★★	C ★★★	B ★★★	B ★★★★★	A ★★	C ★★
Q61	**Q62**	**Q63**	**Q64**	**Q65**	**Q66**	**Q67**	**Q68**	**Q69**	**Q70**
D ★★★	A ★★★★★	A ★★★★★	D ★★★	D ★★★★★	A ★★★★★	C ★★★	A ★★★	C ★★★★★	B ★★★
Q71	**Q72**	**Q73**	**Q74**	**Q75**	**Q76**	**Q77**	**Q78**	**Q79**	**Q80**
B ★★★	C ★★★	A ★★★	D ★★★	B ★★★	A ★★★	B ★★★★★	D ★★★	D ★★★★★	D ★★★
Q81	**Q82**	**Q83**	**Q84**	**Q85**	**Q86**	**Q87**	**Q88**	**Q89**	**Q90**
C ★★★★★	B ★★★★★	B ★★★	A ★★★	B ★★★★★	C ★★★★★	B ★★★★★	D ★★★★★	B ★★★	D ★★★★★

Scoring Guide: Each multiple choice question is worth 1 point (90 total). Each free response question is worth 9 points (90 total). Maximum raw score: 180 points.

Question 01 Analysis

★★ IMPORTANT

Correct Answer: (C) $12(3x + 1)^3$

STEP-BY-STEP SOLUTION

Chain Rule: outer derivative $4(3x+1)^3$ times inner derivative 3 gives $12(3x+1)^3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets to multiply by the inner derivative 3.
✗ **(B)** Uses only the inner derivative 3 and drops the outer factor 4.
✗ **(D)** Does not reduce the exponent from 4 to 3.

Question 02 Analysis

★★ IMPORTANT

Correct Answer: (D) 5 cos(5x)

STEP-BY-STEP SOLUTION

Outer derivative cos(5x) times inner derivative 5 gives 5cos(5x).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Does not change sine to cosine.
✗ **(B)** Forgets the inner derivative 5.
✗ **(C)** Sign error; the derivative of sine is positive cosine.

Question 03 Analysis

★★★ MANDATORY

Correct Answer: (C) $-2x \sin(x^2)$

STEP-BY-STEP SOLUTION

Outer derivative $-\sin(x^2)$ times inner derivative $2x$ gives $-2x \sin(x^2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the inner derivative 2x.
✗ **(B)** Drops the negative sign from the derivative of cosine.
✗ **(D)** Keeps cosine instead of changing to sine.

Question 04 Analysis

★★★ MANDATORY

Correct Answer: (C) 6x·e^(3x²)

STEP-BY-STEP SOLUTION

The derivative of e^u is e^u·u′. Here u = 3x², so u′ = 6x, giving 6x·e^(3x²).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses 3x instead of the correct inner derivative 6x.

✗ **(B)** Changes the exponent of the exponential function incorrectly.

✗ **(D)** Forgets the inner derivative entirely.

Question 05 Analysis

★★★ MANDATORY

Correct Answer: (A) 10x/(5x² + 1)

STEP-BY-STEP SOLUTION

Derivative of ln u is u′/u. Here u′ = 10x, so the result is 10x/(5x² + 1).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses 5x instead of the correct inner derivative 10x.

✗ **(C)** Forgets the inner derivative in the numerator.

✗ **(D)** Keeps only the inner derivative and drops the denominator.

Question 06 Analysis

★★★ MANDATORY

Correct Answer: (D) 3√2 / 4

STEP-BY-STEP SOLUTION

f′(x) = 3x² / (2√(x³+1)). At x = 1: 3/(2√2) = 3√2/4.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the square root and uses 2(x³+1) = 4 in the denominator.

✗ **(B)** Omits the factor 1/2 from the power rule.

✗ **(C)** Treats √(x³+1) as 1 instead of √2.

Question 07 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 3 sec²(3x)

STEP-BY-STEP SOLUTION

Derivative of tan u is $\sec^2 u \cdot u'$. With $u = 3x$, $u' = 3$, giving $3\sec^2(3x)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses the derivative of secant instead of tangent.

✗ **(C)** Squares tangent instead of using secant squared.

✗ **(D)** Forgets the inner derivative 3.

Question 08 Analysis

★★★ MANDATORY

Correct Answer: (D) 0

STEP-BY-STEP SOLUTION

$f'(x) = 5(x^2-4)^4 \cdot 2x = 10x(x^2-4)^4$. At $x = 2$, the factor $(4 - 4) = 0$, so $f'(2) = 0$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates only the factor 10x and ignores $(x^2-4)^4 = 0$.

✗ **(B)** Multiplies 5 by 2^4 while ignoring that the base is zero.

✗ **(C)** Multiplies 10x by 2^3 without noticing the zero factor.

Question 09 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 2 sin x cos x

STEP-BY-STEP SOLUTION

Write $\sin^2 x = (\sin x)^2$. Chain Rule gives $2(\sin x) \cdot \cos x = 2 \sin x \cos x$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Applies the power rule to the wrong function.

✗ **(C)** Forgets the inner derivative cos x.

✗ **(D)** Introduces an incorrect negative sign.

Question 10 Analysis

★★★ MANDATORY

Correct Answer: (D) 0

STEP-BY-STEP SOLUTION

f′(x) = e^(sin x)·cos x. At x = π/2, cos(π/2) = 0, so f′(π/2) = e·0 = 0.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Evaluates e^0 instead of using sin(π/2) = 1.

✗ **(B)** Includes an incorrect negative sign and drops the zero factor.

✗ **(C)** Forgets to multiply by cos(π/2) = 0.

Question 11 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) −1

STEP-BY-STEP SOLUTION

f′(x) = (1/cos x)(−sin x) = −tan x. At x = π/4, −tan(π/4) = −1.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Drops the negative sign from the derivative of cosine.

✗ **(C)** Uses sin(π/4) alone instead of the tangent ratio.

✗ **(D)** Treats ln(cos x) as a constant.

Question 12 Analysis

★★★ MANDATORY

Correct Answer: (C) 2 sec(2x) tan(2x)

STEP-BY-STEP SOLUTION

Derivative of sec u is sec u tan u · u′. With u = 2x, u′ = 2, giving 2sec(2x)tan(2x).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the inner derivative 2.

✗ **(B)** Uses the derivative of tangent instead of secant.

✗ **(D)** Uses an inconsistent argument in the tangent factor.

Question 13 Analysis

★★★ MANDATORY

Correct Answer: (B) cos x / (2√(sin x))

STEP-BY-STEP SOLUTION

Power rule plus chain: (1/2)(sin x)^(−1/2)·cos x = cos x/(2√(sin x)).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Introduces an incorrect negative sign.
✗ **(C)** Forgets the inner derivative cos x.
✗ **(D)** Omits the factor 1/2 from the power rule.

Question 14 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) 12

STEP-BY-STEP SOLUTION

$g'(x) = f'(x^2)\cdot 2x$. So $g'(2) = f'(4)\cdot 4 = 3\cdot 4 = 12$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Multiplies 3 by 3 instead of 4.
✗ **(C)** Reports f′(4) without multiplying by the inner derivative.
✗ **(D)** Uses an inner derivative of 2 instead of 2x = 4.

Question 15 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 10

STEP-BY-STEP SOLUTION

$h'(1) = f'(g(1))\cdot g'(1) = f'(3)\cdot 5 = 2\cdot 5 = 10$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Adds the two derivative values instead of multiplying.
✗ **(C)** Uses an incorrect combination of the given values.
✗ **(D)** Multiplies g(1) by g′(1) rather than using f′(3).

Question 16 Analysis

★★★ MANDATORY

Correct Answer: (A) $-3/(3x + 2)^2$

STEP-BY-STEP SOLUTION

Write as (3x+2)^(−1). Derivative: −(3x+2)^(−2)·3 = −3/(3x+2)².

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Drops the negative sign from the power rule.
✗ **(C)** Divides by 3 instead of multiplying by 3.
✗ **(D)** Forgets the inner derivative 3.

Question 17 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $-6\cos^2(2x)\sin(2x)$

STEP-BY-STEP SOLUTION

Two chain steps: $3\cos^2(2x)\cdot(-\sin(2x))\cdot 2 = -6\cos^2(2x)\sin(2x)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Misses both the factor 2 and the negative sign.
✗ **(C)** Applies only one chain step and misses the factor 2.
✗ **(D)** Drops the negative sign from the derivative of cosine.

Question 18 Analysis

★★★ MANDATORY

Correct Answer: (A) −sin x · e^(cos x)

STEP-BY-STEP SOLUTION

Derivative of e^u is e^u·u′ with u = cos x, so u′ = −sin x, giving −sin x·e^(cos x).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Changes the exponent incorrectly.
✗ **(C)** Drops the negative sign.
✗ **(D)** Forgets the inner derivative.

Question 19 Analysis

★★★ MANDATORY

Correct Answer: (D) 1

STEP-BY-STEP SOLUTION

$f'(x) = 2x/(x^2+1)$. At $x = 1$: $2/2 = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the inner derivative 2x in the numerator.

✗ **(B)** Treats the logarithm as a constant.

✗ **(C)** Uses only the numerator 2x and ignores the denominator.

Question 20 Analysis

★★★ MANDATORY

Correct Answer: (A) cos(ln x) / x

STEP-BY-STEP SOLUTION

Outer derivative cos(ln x) times inner derivative 1/x gives cos(ln x)/x.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Places the cosine factor in the denominator.

✗ **(C)** Forgets the inner derivative 1/x.

✗ **(D)** Uses the derivative of cosine instead of sine.

Question 21 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 0

STEP-BY-STEP SOLUTION

$f'(x) = 3(x + 1/x)^2(1 - 1/x^2)$. At $x = 1$, the factor $(1 - 1) = 0$, so $f'(1) = 0$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Doubles an already incorrect value.

✗ **(B)** Computes $3(2)^2$ and ignores the inner derivative.

✗ **(D)** Uses only the leading coefficient 3 and ignores the zero factor.

Question 22 Analysis

★★★ MANDATORY

Correct Answer: (B) 3 ln 2 · 2^(3x)

STEP-BY-STEP SOLUTION

Derivative of a^u is a^u (ln a) u′. With a = 2 and u = 3x, the result is 3 ln2 · 2^(3x).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the factor ln 2.
✗ **(C)** Forgets the inner derivative 3.
✗ **(D)** Incorrectly applies the power rule to an exponential function.

Question 23 Analysis

★★★ MANDATORY

Correct Answer: (D) $2x \sec^2(x^2)$

STEP-BY-STEP SOLUTION

Derivative of tan u is $\sec^2 u \cdot u'$. With $u = x^2$, $u' = 2x$, giving $2x \sec^2(x^2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Changes the argument from x^2 to $2x$.
✗ **(B)** Forgets the inner derivative $2x$.
✗ **(C)** Squares tangent instead of using secant squared.

Question 24 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $-\sqrt{3}/3$

STEP-BY-STEP SOLUTION

$f'(x) = -x/\sqrt{(4 - x^2)}$. At $x = 1$: $-1/\sqrt{3} = -\sqrt{3}/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Drops the negative sign coming from the inner derivative $-2x$.
✗ **(C)** Omits the factor 1/2 from the power rule.
✗ **(D)** Treats $\sqrt{3}$ as 3 in the denominator.

Question 25 Analysis

★★★ MANDATORY

Correct Answer: (A) $(3x^2 + 2)/(x^3 + 2x)$

STEP-BY-STEP SOLUTION

Derivative of ln u is u′/u with u′ = $3x^2 + 2$, giving $(3x^2+2)/(x^3+2x)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Omits the constant term 2 when differentiating the inside.

✗ **(C)** Forgets the inner derivative in the numerator.

✗ **(D)** Keeps only the inner derivative and drops the denominator.

Question 26 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) −x/y

STEP-BY-STEP SOLUTION

Differentiate: 2x + 2y·y′ = 0, so y′ = −x/y.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Inverts the ratio of x and y.

✗ **(B)** Inverts the ratio and keeps the sign.

✗ **(D)** Drops the negative sign when solving for y′.

Question 27 Analysis

★★★ MANDATORY

Correct Answer: (A) −3/4

STEP-BY-STEP SOLUTION

Since y′ = −x/y, at (3, 4) the slope is −3/4.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses the reciprocal of the correct slope.

✗ **(C)** Drops the negative sign.

✗ **(D)** Uses the reciprocal without the negative sign.

Question 28 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) −y/x

STEP-BY-STEP SOLUTION

Product rule: $y + x \cdot y' = 0$, so $y' = -y/x$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Inverts the ratio and keeps the wrong sign placement.

✗ **(C)** Inverts the ratio of x and y.

✗ **(D)** Drops the negative sign.

Question 29 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) −(2x + y)/x

STEP-BY-STEP SOLUTION

Differentiate: $2x + (y + x \cdot y') = 0$, so $x \cdot y' = -(2x + y)$ and $y' = -(2x + y)/x$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the product rule on the xy term.

✗ **(B)** Drops the negative sign when isolating y′.

✗ **(D)** Mixes up which terms belong in the numerator.

Question 30 Analysis

★★★ MANDATORY

Correct Answer: (C) $3x^2/(2y)$

STEP-BY-STEP SOLUTION

Differentiate: $2y \cdot y' = 3x^2$, so $y' = 3x^2/(2y)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Introduces an incorrect negative sign.

✗ **(B)** Forgets the factor 2 from differentiating y^2.

✗ **(D)** Inverts the resulting fraction.

Question 31 Analysis

★★★ MANDATORY

Correct Answer: (B) 1/cos y

STEP-BY-STEP SOLUTION

Differentiate: cos y · y′ = 1, so y′ = 1/cos y.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses sine instead of cosine in the denominator.
✗ **(C)** Forgets to solve for y′ after differentiating.
✗ **(D)** Introduces an incorrect negative sign.

Question 32 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 2/x

STEP-BY-STEP SOLUTION

Differentiate: e^y·y′ = 2x, so y′ = 2x/e^y. Since e^y = x^2, this becomes $2x/x^2 = 2/x$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Inverts the final simplified fraction.
✗ **(C)** Forgets to divide by e^y.
✗ **(D)** Multiplies by e^y instead of dividing.

Question 33 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) −1/4

STEP-BY-STEP SOLUTION

$3x^2 + 3y^2 \cdot y' = 0$ gives $y' = -x^2/y^2$. At (1, 2): −1/4.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets to square the y-value.
✗ **(C)** Inverts the fraction.
✗ **(D)** Drops the negative sign.

Question 34 Analysis

★★★ MANDATORY

Correct Answer: (A) −1

STEP-BY-STEP SOLUTION

Product rule: $2xy + x^2 \cdot y' = 0$, so $y' = -2y/x$. At (2, 1): $-2/2 = -1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets to divide by x after isolating y′.

✗ **(C)** Inverts the resulting fraction.

✗ **(D)** Drops the negative sign.

Question 35 Analysis

★★★ MANDATORY

Correct Answer: (B) (0, 5) and (0, −5)

STEP-BY-STEP SOLUTION

$y' = -x/y$ equals 0 when $x = 0$, giving $y = \pm 5$, so the points are (0, 5) and (0, −5).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** This point has a nonzero slope of −3/4.

✗ **(C)** These are the points of vertical tangency, where y = 0.

✗ **(D)** The origin is not on the circle.

Question 36 Analysis

★★★ MANDATORY

Correct Answer: (B) (5, 0) and (−5, 0)

STEP-BY-STEP SOLUTION

$y' = -x/y$ is undefined when $y = 0$, giving $x = \pm 5$, so the points are (5, 0) and (−5, 0).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** A circle does have vertical tangents at its leftmost and rightmost points.

✗ **(C)** This point has a finite nonzero slope.

✗ **(D)** These are the points of horizontal tangency.

Question 37 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $-1/y^3$

STEP-BY-STEP SOLUTION

First $y' = -x/y$. Then $y'' = -[y - x \cdot y']/y^2 = -[y + x^2/y]/y^2 = -(x^2 + y^2)/y^3 = -1/y^3$ using $x^2 + y^2 = 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the wrong power of y in the denominator.

✗ **(B)** Drops the negative sign.

✗ **(C)** This is the first derivative, not the second.

Question 38 Analysis

★★★ MANDATORY

Correct Answer: (B) 0

STEP-BY-STEP SOLUTION

$8x + 18y \cdot y' = 0$ gives $y' = -4x/(9y)$. At $(0, 2)$: $y' = 0$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses the coordinates directly instead of the derivative.

✗ **(C)** Ignores the x-value of 0 in the numerator.

✗ **(D)** The denominator $9y = 18$ is nonzero, so the slope is defined.

Question 39 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $y/(y - 1)$

STEP-BY-STEP SOLUTION

Differentiate: $y'/y + 1 = y'$, so $1 = y'(1 - 1/y) = y'(y-1)/y$, giving $y' = y/(y-1)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Loses the factor y in the numerator.

✗ **(C)** Reverses the sign in the denominator.

✗ **(D)** Inverts the correct expression.

Question 40 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) y cos(xy) / (1 – x cos(xy))

STEP-BY-STEP SOLUTION

$y' = \cos(xy)(y + x \cdot y')$. Expanding: $y' - x\cos(xy) \cdot y' = y\cos(xy)$, so $y' = y\cos(xy)/(1 - x\cos(xy))$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the product rule on the inside and never solves for y′.

✗ **(C)** Mixes up the roles of x and y in the expression.

✗ **(D)** Stops before collecting the y′ terms.

Question 41 Analysis

★★★ MANDATORY

Correct Answer: (C) −y/(x + 2y)

STEP-BY-STEP SOLUTION

$y + x \cdot y' + 2y \cdot y' = 0$, so $y'(x + 2y) = -y$ and $y' = -y/(x + 2y)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Drops the negative sign.

✗ **(B)** Forgets the factor 2 from differentiating y^2.

✗ **(D)** Places x in the numerator instead of y.

Question 42 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $1/(3y^2 + 1)$

STEP-BY-STEP SOLUTION

Differentiate with respect to x: $1 = (3y^2 + 1) \cdot y'$, so $y' = 1/(3y^2 + 1)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Omits the constant term 1 from the inner derivative.

✗ **(C)** Introduces an incorrect negative sign.

✗ **(D)** This is dx/dy, not dy/dx.

Question 43 Analysis

★★★ MANDATORY

Correct Answer: (A) $y - 4 = -(3/4)(x - 3)$

STEP-BY-STEP SOLUTION

The slope is $-x/y = -3/4$, and the point is $(3, 4)$, giving $y - 4 = -(3/4)(x - 3)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Omits the point entirely and passes through the origin.
✗ **(C)** Drops the negative sign on the slope.
✗ **(D)** Swaps the x- and y-coordinates of the point.

Question 44 Analysis

★★★ MANDATORY

Correct Answer: (A) $-1/\sin y$

STEP-BY-STEP SOLUTION

Differentiate: $-\sin y \cdot y' = 1$, so $y' = -1/\sin y$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets to solve for y'.
✗ **(C)** Uses cosine instead of sine in the denominator.
✗ **(D)** Drops the negative sign.

Question 45 Analysis

★★★ MANDATORY

Correct Answer: (A) 5/3

STEP-BY-STEP SOLUTION

$2x - 2y \cdot y' = 0$ gives $y' = x/y$. At $(5, 3)$: $5/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Inverts the fraction and adds a wrong sign.
✗ **(C)** Applies the sign rule for a circle instead of a hyperbola.
✗ **(D)** Inverts the correct fraction.

Question 46 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) −1/x

STEP-BY-STEP SOLUTION

Product rule: e^y + x e^y y′ = 0. Factoring e^y (never zero) gives 1 + x y′ = 0, so y′ = −1/x.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Does not cancel e^y from numerator and denominator.

✗ **(B)** Fails to cancel the common factor e^y.

✗ **(C)** Drops the negative sign.

Question 47 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $1/f'(f^{-1}(b))$

STEP-BY-STEP SOLUTION

The inverse function derivative rule is $(f^{-1})'(b) = 1 / f'(f^{-1}(b))$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Adds an incorrect negative sign.

✗ **(C)** Omits the reciprocal entirely.

✗ **(D)** Evaluates f′ at b instead of at $f^{-1}(b)$.

Question 48 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 1/4

STEP-BY-STEP SOLUTION

Since f(1) = 3, g(3) = 1. Also $f'(x) = 3x^2 + 1$, so f′(1) = 4. Thus g′(3) = 1/4.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Reports the input value instead of the derivative.

✗ **(C)** Omits the reciprocal.

✗ **(D)** Evaluates f′ at 3 instead of at g(3) = 1.

Question 49 Analysis

★★★ MANDATORY

Correct Answer: (A) 1/7

STEP-BY-STEP SOLUTION

$f(1) = 2$, so $f^{-1}(2) = 1$. Also $f'(x) = 5x^4 + 2$, so $f'(1) = 7$. Thus $(f^{-1})'(2) = 1/7$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Uses an incorrect value for $f'(1)$.

✗ **(C)** Uses the input value 2 rather than $f'(1)$.

✗ **(D)** Omits the reciprocal.

Question 50 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 1/3

STEP-BY-STEP SOLUTION

Since $f(2) = 5$, $f^{-1}(5) = 2$. Then $(f^{-1})'(5) = 1/f'(2) = 1/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Reports the output value of f rather than a derivative.

✗ **(B)** Omits the reciprocal.

✗ **(C)** Uses the function value 5 instead of the derivative value 3.

Question 51 Analysis

★★★ MANDATORY

Correct Answer: (D) 1/e

STEP-BY-STEP SOLUTION

$g(x) = \ln x$, so $g'(x) = 1/x$ and $g'(e) = 1/e$. Equivalently, $1/f'(1) = 1/e$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Squares the denominator unnecessarily.

✗ **(B)** Evaluates the derivative at the wrong point.

✗ **(C)** Omits the reciprocal.

Question 52 Analysis

★★★ MANDATORY

Correct Answer: (C) 4

STEP-BY-STEP SOLUTION

$g(x) = x^2$, so $g'(x) = 2x$ and $g'(2) = 4$. Check: $1/f'(4) = 1/(1/4) = 4$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Inverts the correct answer.

✗ **(B)** Uses f' directly without taking the reciprocal.

✗ **(D)** Reports the input value instead of the derivative.

Question 53 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 1/3

STEP-BY-STEP SOLUTION

$f(0) = 0$, so $f^{-1}(0) = 0$. Also $f'(x) = 2 + \cos x$, so $f'(0) = 3$. Thus $(f^{-1})'(0) = 1/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses only the derivative of sin x.

✗ **(C)** Omits the reciprocal.

✗ **(D)** Ignores the cosine term when computing $f'(0)$.

Question 54 Analysis

★★★ MANDATORY

Correct Answer: (B) 2

STEP-BY-STEP SOLUTION

$(f^{-1})'(7) = 1/f'(3) = 1/(1/2) = 2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Takes the reciprocal of the wrong quantity.

✗ **(C)** Reports the function value instead of a derivative.

✗ **(D)** Omits the reciprocal step.

Question 55 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 1/5

STEP-BY-STEP SOLUTION

From the table $f(2) = 7$, so $g(7) = 2$. Then $g'(7) = 1/f'(2) = 1/5$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the reciprocal.
✗ **(B)** Uses $f'(1)$ instead of $f'(2)$.
✗ **(D)** Uses $f'(3)$ instead of $f'(2)$.

Question 56 Analysis

★★★ MANDATORY

Correct Answer: (C) 1/12

STEP-BY-STEP SOLUTION

$f(2) = 10$, so $f^{-1}(10) = 2$. Also $f'(x) = 3x^2$, so $f'(2) = 12$. Thus $(f^{-1})'(10) = 1/12$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the reciprocal.
✗ **(B)** Uses only the exponent 3 in the calculation.
✗ **(D)** Uses the input value 10 instead of $f'(2)$.

Question 57 Analysis

★★★ MANDATORY

Correct Answer: (B) positive for all x in the domain of g

STEP-BY-STEP SOLUTION

Since $g'(x) = 1/f'(g(x))$ and f' is always positive, the reciprocal is also always positive.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** A positive reciprocal cannot be negative.
✗ **(C)** g' is defined wherever f' is nonzero.
✗ **(D)** A reciprocal of a finite nonzero number is never zero.

Question 58 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 1/2

STEP-BY-STEP SOLUTION

$f(\pi/4) = 1$, so $g(1) = \pi/4$. Also $f'(x) = \sec^2 x$, so $f'(\pi/4) = 2$. Thus $g'(1) = 1/2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the reciprocal.

✗ **(C)** Uses $\cos(\pi/4)$ rather than the secant squared value.

✗ **(D)** Uses $\sec(\pi/4)$ instead of $\sec^2(\pi/4)$.

Question 59 Analysis

★★ IMPORTANT

Correct Answer: (A) $1/\sqrt{(1 - x^2)}$

STEP-BY-STEP SOLUTION

The standard result is $d/dx \arcsin x = 1/\sqrt{(1 - x^2)}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** This is the derivative of $\arccos x$.

✗ **(C)** This is the derivative of $\arctan x$.

✗ **(D)** This is the negative of the arctan derivative.

Question 60 Analysis

★★ IMPORTANT

Correct Answer: (C) $1/(1 + x^2)$

STEP-BY-STEP SOLUTION

The standard result is $d/dx \arctan x = 1/(1 + x^2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** The denominator sign is incorrect.

✗ **(B)** This is the derivative of $\operatorname{arccot} x$.

✗ **(D)** This is the derivative of $\arcsin x$.

Question 61 Analysis

★★★ MANDATORY

Correct Answer: (D) $-1/\sqrt{(1 - x^2)}$

STEP-BY-STEP SOLUTION

The standard result is d/dx arccos x = $-1/\sqrt{(1 - x^2)}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** The radicand is reversed.

✗ **(B)** This is the derivative of arcsin x; the sign is missing.

✗ **(C)** This is the derivative of arccot x.

Question 62 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $2/(1 + 4x^2)$

STEP-BY-STEP SOLUTION

Apply the chain rule: $(1/(1 + (2x)^2))\cdot 2 = 2/(1 + 4x^2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Does not square the coefficient 2 in the denominator.

✗ **(C)** Forgets the inner derivative 2.

✗ **(D)** Fails to square the inner function in the denominator.

Question 63 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $2x/\sqrt{(1 - x^4)}$

STEP-BY-STEP SOLUTION

Chain rule: $(1/\sqrt{(1 - (x^2)^2)})\cdot 2x = 2x/\sqrt{(1 - x^4)}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Does not substitute x^2 into the radicand.

✗ **(C)** Uses the arctan form instead of arcsin.

✗ **(D)** Forgets the inner derivative 2x.

Question 64 Analysis

★★★ MANDATORY

Correct Answer: (D) 1/2

STEP-BY-STEP SOLUTION

$f'(x) = 1/(1 + x^2)$. At $x = 1$: 1/2.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Ignores the x^2 term in the denominator.

✗ **(B)** Inverts the correct value.

✗ **(C)** Reports the function value arctan 1 instead of the derivative.

Question 65 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $1/\sqrt{(9 - x^2)}$

STEP-BY-STEP SOLUTION

$(1/\sqrt{(1 - x^2/9)})\cdot(1/3) = (3/\sqrt{(9 - x^2)})\cdot(1/3) = 1/\sqrt{(9 - x^2)}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses 3x instead of x/3 inside the radical.

✗ **(B)** Does not clear the fraction inside the radical correctly.

✗ **(C)** Forgets to multiply by the inner derivative 1/3.

Question 66 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (A) $\arctan x + x/(1 + x^2)$

STEP-BY-STEP SOLUTION

Product rule: $(1)(\arctan x) + x\cdot(1/(1 + x^2)) = \arctan x + x/(1 + x^2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Omits the first term of the product rule.

✗ **(C)** Forgets the factor x in the second term.

✗ **(D)** Uses only the derivative of arctan x and skips the product rule.

Question 67 Analysis

★★★ MANDATORY

Correct Answer: (C) $-2/\sqrt{(1-4x^2)}$

STEP-BY-STEP SOLUTION

$(-1/\sqrt{(1-(2x)^2)})\cdot 2 = -2/\sqrt{(1-4x^2)}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Drops the negative sign of the arccos derivative.
✗ **(B)** Does not square the coefficient 2 in the radicand.
✗ **(D)** Forgets the inner derivative 2.

Question 68 Analysis

★★★ MANDATORY

Correct Answer: (A) e^(arctan x)/(1 + x^2)

STEP-BY-STEP SOLUTION

Derivative of e^u is e^u·u′ with u = arctan x, so u′ = $1/(1+x^2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Forgets the inner derivative.
✗ **(C)** Differentiates inside the exponent instead of multiplying.
✗ **(D)** Misapplies the exponential derivative rule.

Question 69 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) $-1/(1+x^2)$

STEP-BY-STEP SOLUTION

$(1/(1+1/x^2))\cdot(-1/x^2) = (x^2/(x^2+1))\cdot(-1/x^2) = -1/(1+x^2)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Forgets the negative inner derivative.
✗ **(B)** Uses only the inner derivative and drops the outer factor.
✗ **(D)** Fails to simplify the compound fraction.

Question 70 Analysis

★★★ MANDATORY

Correct Answer: (B) $e^x/\sqrt{(1 - e^{(2x)})}$

STEP-BY-STEP SOLUTION

$(1/\sqrt{(1 - (e^x)^2)})\cdot e^x = e^x/\sqrt{(1 - e^{(2x)})}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Does not square e^x inside the radical.

✗ **(C)** Forgets the inner derivative e^x.

✗ **(D)** Uses the arctan form instead of arcsin.

Question 71 Analysis

★★★ MANDATORY

Correct Answer: (B) $2x/(1 + x^4)$

STEP-BY-STEP SOLUTION

$(1/(1 + (x^2)^2))\cdot 2x = 2x/(1 + x^4)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Does not square x^2 in the denominator.

✗ **(C)** Forgets the inner derivative $2x$.

✗ **(D)** Uses the arcsin form instead of arctan.

Question 72 Analysis

★★★ MANDATORY

Correct Answer: (C) $2\sqrt{3}/3$

STEP-BY-STEP SOLUTION

$f'(x) = 1/\sqrt{(1 - x^2)}$. At $x = 1/2$: $1/\sqrt{(3/4)} = 2/\sqrt{3} = 2\sqrt{3}/3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Reports the input value instead of the derivative.

✗ **(B)** Inverts part of the radical incorrectly.

✗ **(D)** Forgets to take the square root of 3/4.

Question 73 Analysis

★★★ MANDATORY

Correct Answer: (A) 3

STEP-BY-STEP SOLUTION

$f'(x) = 3/(1 + 9x^2)$. At $x = 0$: $3/1 = 3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Assumes the derivative is zero because arctan 0 = 0.
✗ **(C)** Inverts the correct answer.
✗ **(D)** Forgets the inner derivative 3.

Question 74 Analysis

★★★ MANDATORY

Correct Answer: (D) 24x

STEP-BY-STEP SOLUTION

$f' = 4x^3$, $f'' = 12x^2$, $f''' = 24x$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the second derivative.
✗ **(B)** This is the first derivative.
✗ **(C)** This is the fourth derivative.

Question 75 Analysis

★★★ MANDATORY

Correct Answer: (B) –sin x

STEP-BY-STEP SOLUTION

$f' = \cos x$ and $f'' = -\sin x$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the first derivative.
✗ **(C)** This is the third derivative.
✗ **(D)** This is the fourth derivative.

Question 76 Analysis

★★★ MANDATORY

Correct Answer: (A) 4e^(2x)

STEP-BY-STEP SOLUTION

f′ = 2e^(2x) and f″ = 4e^(2x).

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Ignores the chain rule entirely.

✗ **(C)** Changes the exponent incorrectly.

✗ **(D)** This is the first derivative.

Question 77 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $-1/x^2$

STEP-BY-STEP SOLUTION

$f' = 1/x = x^{-1}$, so $f'' = -x^{-2} = -1/x^2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the third derivative of ln x.

✗ **(C)** Drops the negative sign from the power rule.

✗ **(D)** This is the first derivative.

Question 78 Analysis

★★★ MANDATORY

Correct Answer: (D) $2/x^3$

STEP-BY-STEP SOLUTION

$f' = -x^{-2}$, so $f'' = 2x^{-3} = 2/x^3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses an incorrect sign for the second derivative.

✗ **(B)** Drops the sign of the first derivative.

✗ **(C)** This is the first derivative.

Question 79 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) −6

STEP-BY-STEP SOLUTION

$v(t) = 3t^2 - 12t$ and $a(t) = 6t - 12$. At $t = 1$: $a(1) = -6$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is the velocity $v(1)$, not the acceleration.

✗ **(B)** Drops the constant term −12.

✗ **(C)** Uses an incorrect coefficient when differentiating.

Question 80 Analysis

★★★ MANDATORY

Correct Answer: (D) 10

STEP-BY-STEP SOLUTION

$f' = 3x^2 + 4x$ and $f'' = 6x + 4$. At $x = 1$: $f''(1) = 10$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** This is $f'(1)$, not $f''(1)$.

✗ **(B)** Omits the constant term 4 from f''.

✗ **(C)** Uses only the constant term of f''.

Question 81 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) −4 cos(2x)

STEP-BY-STEP SOLUTION

First derivative: $-2\sin(2x)$. Second derivative: $-4\cos(2x)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Drops the negative sign from the second differentiation.

✗ **(B)** This is the first derivative.

✗ **(D)** Omits the factor 4 from the two chain rule applications.

Question 82 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) $2x \sin(3x) + 3x^2 \cos(3x)$

STEP-BY-STEP SOLUTION

Product rule with chain rule: $2x \cdot \sin(3x) + x^2 \cdot 3\cos(3x)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the first term of the product rule.

✗ **(C)** Multiplies the derivatives instead of using the product rule.

✗ **(D)** Forgets the inner derivative 3 in the second term.

Question 83 Analysis

★★★ MANDATORY

Correct Answer: (B) $x \cdot e^x/(x + 1)^2$

STEP-BY-STEP SOLUTION

Quotient rule: $[e^x(x+1) - e^x(1)]/(x+1)^2 = e^x \cdot x/(x+1)^2$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Reverses the order of terms in the numerator.

✗ **(C)** Fails to simplify the numerator correctly.

✗ **(D)** Adds instead of subtracting in the numerator.

Question 84 Analysis

★★★ MANDATORY

Correct Answer: (A) $(\ln x)/(2\sqrt{x}) + 1/\sqrt{x}$

STEP-BY-STEP SOLUTION

Product rule: $(1/(2\sqrt{x})) \cdot \ln x + \sqrt{x} \cdot (1/x) = (\ln x)/(2\sqrt{x}) + 1/\sqrt{x}$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(B)** Multiplies the two derivatives instead of using the product rule.

✗ **(C)** Omits the second term of the product rule.

✗ **(D)** Combines the terms over an incorrect common denominator.

Question 85 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 15

STEP-BY-STEP SOLUTION

$h'(2) = f'(g(2))\cdot g'(2) = f'(1)\cdot 5 = 3\cdot 5 = 15$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses $f'(2) = 7$ instead of $f'(1) = 3$.

✗ **(C)** Reports $f'(2)$ without applying the chain rule.

✗ **(D)** Uses $g'(1) = 6$ instead of $g'(2) = 5$ in an incorrect product.

Question 86 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (C) 42

STEP-BY-STEP SOLUTION

$k'(2) = g'(f(2))\cdot f'(2) = g'(1)\cdot 7 = 6\cdot 7 = 42$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Uses $f'(1) = 3$ instead of $f'(2) = 7$.

✗ **(B)** Uses $g'(2) = 5$ instead of $g'(1) = 6$.

✗ **(D)** Computes the chain rule in the reverse order.

Question 87 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (B) 1/x + 1

STEP-BY-STEP SOLUTION

Simplify first: $\ln(x\cdot e^x) = \ln x + x$. The derivative is $1/x + 1$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Differentiates the simplified expression incorrectly.

✗ **(C)** Differentiates only the logarithm of x.

✗ **(D)** Applies the quotient rule where it does not belong.

Question 88 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) 2x

STEP-BY-STEP SOLUTION

Simplify: $e^{(2 \ln x)} = e^{(\ln x^2)} = x^2$. The derivative is $2x$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Differentiates only the exponent.

✗ **(B)** Applies the chain rule without simplifying and then drops a factor.

✗ **(C)** Reports the simplified function rather than its derivative.

Question 89 Analysis

★★★ MANDATORY

Correct Answer: (B) $6(2x + 1)^2(x - 1) + (2x + 1)^3$

STEP-BY-STEP SOLUTION

Product rule with chain rule: $[3(2x+1)^2 \cdot 2](x-1) + (2x+1)^3(1) = 6(2x+1)^2(x-1) + (2x+1)^3$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Omits the second term of the product rule.

✗ **(C)** Forgets the inner derivative 2 in the first term.

✗ **(D)** Uses neither the product rule nor the chain rule correctly.

Question 90 Analysis

★★★★★ VERY IMPORTANT

Correct Answer: (D) $-\sin x \cdot \cos(\cos x)$

STEP-BY-STEP SOLUTION

Outer derivative $\cos(\cos x)$ times inner derivative $-\sin x$ gives $-\sin x \cdot \cos(\cos x)$.

WHY OTHER OPTIONS ARE INCORRECT

✗ **(A)** Drops the negative sign from the derivative of cosine.

✗ **(B)** Forgets the inner derivative.

✗ **(C)** Swaps the inner and outer functions.

Question 91 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) By the Chain Rule, $h'(x) = f'(g(x))\cdot g'(x)$.
$h'(1) = f'(g(1))\cdot g'(1) = f'(2)\cdot(-4) = 7\cdot(-4) = \mathbf{-28}$.

(b) By the Chain Rule, $k'(x) = g'(f(x))\cdot f'(x)$.
$k'(1) = g'(f(1))\cdot f'(1) = g'(3)\cdot 5 = 8\cdot 5 = \mathbf{40}$.

(c) By the Product Rule, $m'(x) = f'(x)g(x) + f(x)g'(x)$.
$m'(3) = f'(3)\cdot g(3) + f(3)\cdot g'(3) = (-2)(1) + (2)(8) = -2 + 16 = \mathbf{14}$.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct Chain Rule setup $f'(g(1))\cdot g'(1)$	2 pts
Part (a): Correct value -28	1 pt
Part (b): Correct Chain Rule setup $g'(f(1))\cdot f'(1)$	2 pts
Part (b): Correct value 40	1 pt
Part (c): Correct Product Rule setup	2 pts
Part (c): Correct value 14	1 pt

Question 92 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) Differentiate both sides with respect to x, using the Product Rule on xy:
$2x + (y + x \cdot y') + 2y \cdot y' = 0$
$y'(x + 2y) = -(2x + y)$
$dy/dx = -(2x + y)/(x + 2y)$.

(b) At (1, 2): $dy/dx = -(2 + 2)/(1 + 4) = -4/5$.
Tangent line: **$y - 2 = -(4/5)(x - 1)$**.

(c) A horizontal tangent requires $2x + y = 0$, so $y = -2x$ (with $x + 2y \neq 0$).
Substitute into the curve: $x^2 + x(-2x) + (-2x)^2 = 7 \Rightarrow x^2 - 2x^2 + 4x^2 = 3x^2 = 7 \Rightarrow x = \pm\sqrt{(7/3)} = \pm\sqrt{21}/3$.
Check $x + 2y = x - 4x = -3x \neq 0$ since $x \neq 0$.
Points: **$(\sqrt{21}/3, -2\sqrt{21}/3)$ and $(-\sqrt{21}/3, 2\sqrt{21}/3)$**.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct implicit differentiation including Product Rule on xy	2 pts
Part (a): Correct algebra isolating dy/dx	1 pt
Part (b): Correct slope −4/5 at (1, 2)	2 pts
Part (b): Correct tangent line equation	1 pt
Part (c): Sets numerator 2x + y = 0 and substitutes	2 pts
Part (c): Correct coordinates of both points	1 pt

Question 93 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) $f'(x) = 3x^2 + 2$. Since $3x^2 \geq 0$, we have $f'(x) \geq 2 > 0$ for all x.
Because $f'(x) > 0$ everywhere, f is strictly increasing, so f is one-to-one and therefore has an inverse.

(b) $f(1) = 1 + 2 - 1 = 2$, so **$g(2) = 1$**.
By the inverse derivative rule, $g'(2) = 1/f'(g(2)) = 1/f'(1)$.
$f'(1) = 3(1)^2 + 2 = 5$, so **$g'(2) = 1/5$**.

(c) The point on $y = g(x)$ is $(2, 1)$ and the slope is $1/5$.
Tangent line: **$y - 1 = (1/5)(x - 2)$**.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Computes $f'(x) = 3x^2 + 2$ and argues it is positive	1 pt
Part (a): Concludes f is one-to-one and invertible	1 pt
Part (b): Finds $g(2) = 1$ by solving $f(x) = 2$	2 pts
Part (b): Applies $g'(2) = 1/f'(1)$ to obtain $1/5$	2 pts
Part (c): Identifies the point $(2, 1)$	1 pt
Part (c): Correct tangent line equation	2 pts

Question 94 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) $h'(x) = (1/(1 + (x^2)^2)) \cdot 2x = \mathbf{2x/(1 + x^4)}$.
$h'(1) = 2/(1 + 1) = \mathbf{1}$.

(b) $p'(x) = (1/\sqrt{(1 - (3x)^2)}) \cdot 3 = \mathbf{3/\sqrt{(1 - 9x^2)}}$.
This requires $1 - 9x^2 > 0$, so $x^2 < 1/9$. The interval is $\mathbf{(-1/3, 1/3)}$.

(c) By the Product Rule: $q'(x) = (1)(\arccos x) + x \cdot (-1/\sqrt{(1 - x^2)}) = \mathbf{\arccos x - x/\sqrt{(1 - x^2)}}$.
$q'(0) = \arccos 0 - 0 = \mathbf{\pi/2}$.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct chain rule giving $2x/(1 + x^4)$	2 pts
Part (a): Correct value $h'(1) = 1$	1 pt
Part (b): Correct derivative $3/\sqrt{(1 - 9x^2)}$	2 pts
Part (b): Correct interval $(-1/3, 1/3)$	1 pt
Part (c): Correct Product Rule with arccos derivative	2 pts
Part (c): Correct value $\pi/2$	1 pt

Question 95 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) $v(t) = s'(t) = 3t^2 - 12t + 9 = \mathbf{3(t-1)(t-3)}$.

(b) $a(t) = v'(t) = \mathbf{6t - 12}$. Therefore $a(2) = 12 - 12 = \mathbf{0}$.

(c) The particle is at rest when $v(t) = 0$, which gives $\mathbf{t = 1}$ **and** $\mathbf{t = 3}$.

(d) $v(2.5) = 3(1.5)(-0.5) = -2.25 < 0$, and $a(2.5) = 15 - 12 = 3 > 0$.
Since velocity and acceleration have **opposite signs**, the speed is **decreasing** at $t = 2.5$.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct velocity and factored form	2 pts
Part (b): Correct acceleration and $a(2) = 0$	2 pts
Part (c): Correct times $t = 1$ and $t = 3$	2 pts
Part (d): Computes signs of $v(2.5)$ and $a(2.5)$	2 pts
Part (d): Correct conclusion that speed is decreasing	1 pt

Question 96 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) Product Rule with Chain Rule on each factor:
f′(x) = 2e^(2x)·sin(3x) + e^(2x)·3cos(3x) = **e^(2x)(2 sin 3x + 3 cos 3x)**.

(b) f′(0) = e⁰(2·0 + 3·1) = **3**.

(c) Differentiate f′(x) = e^(2x)(2 sin 3x + 3 cos 3x) using the Product Rule:
f″(x) = 2e^(2x)(2 sin 3x + 3 cos 3x) + e^(2x)(6 cos 3x − 9 sin 3x)
= e^(2x)(4 sin 3x + 6 cos 3x + 6 cos 3x − 9 sin 3x) = **e^(2x)(12 cos 3x − 5 sin 3x)**.
f″(0) = 1·(12 − 0) = **12**.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct Product Rule structure	1 pt
Part (a): Correct chain factors 2 and 3	2 pts
Part (b): Correct value f′(0) = 3	2 pts
Part (c): Correct second application of Product Rule	2 pts
Part (c): Correct simplification and f″(0) = 12	2 pts

Question 97 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) Differentiate: $2x + 2y \cdot y' = 4$, so $2y \cdot y' = 4 - 2x$ and **$dy/dx = (2 - x)/y$**.

(b) Horizontal tangent requires the numerator zero and $y \neq 0$: $2 - x = 0$, so $x = 2$.
Substituting: $4 + y^2 = 8 \Rightarrow y^2 = 4 \Rightarrow y = \pm 2$.
Points: **(2, 2) and (2, −2)**.

(c) Vertical tangent requires $y = 0$ (denominator zero) with numerator nonzero.
Substituting $y = 0$: $x^2 = 4x \Rightarrow x(x - 4) = 0 \Rightarrow x = 0$ or $x = 4$.
Points: **(0, 0) and (4, 0)**.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct implicit differentiation	2 pts
Part (a): Correct solved form (2 − x)/y	1 pt
Part (b): Sets numerator equal to zero and substitutes	2 pts
Part (b): Correct points (2, ±2)	1 pt
Part (c): Sets denominator equal to zero and substitutes	2 pts
Part (c): Correct points (0, 0) and (4, 0)	1 pt

Question 98 Official Model Solution

★★★ MANDATORY

COMPREHENSIVE ANALYTICAL SOLUTION

(a) By the Chain Rule, $N'(t) = 1200 \cdot e^{(0.3t)} \cdot 0.3 = \mathbf{360 \cdot e^{(0.3t)}}$.

(b) $N'(2) = 360 \cdot e^{(0.6)} \approx 360(1.8221) \approx \mathbf{655.97}$ **bacteria per hour**.
Interpretation: at time $t = 2$ hours, the number of bacteria is increasing at a rate of about 656 bacteria per hour.

(c) Set $N'(t) = 1080$:
$360 \cdot e^{(0.3t)} = 1080 \Rightarrow e^{(0.3t)} = 3 \Rightarrow 0.3t = \ln 3 \Rightarrow t = (\ln 3)/0.3 \approx \mathbf{3.662}$ **hours**.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct Chain Rule giving $360e^{(0.3t)}$	2 pts
Part (b): Correct numerical value of $N'(2)$	1 pt
Part (b): Correct interpretation with units	2 pts
Part (c): Sets up the equation $360e^{(0.3t)} = 1080$	2 pts
Part (c): Correct solution $t \approx 3.662$ hours	2 pts

Question 99 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) The table shows $f'(x) > 0$ at every listed value, and f is stated to be increasing on $[0, 3]$.
An increasing function is one-to-one, so f has an inverse function g on this interval.

(b) From the table, $f(2) = 7$, so **$g(7) = 2$**.

(c) By the inverse derivative rule, $g'(7) = 1/f'(g(7)) = 1/f'(2) =$ **$1/5$**.

(d) The point is $(7, 2)$ and the slope is $1/5$.
Tangent line: **$y - 2 = (1/5)(x - 7)$**.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): States f is increasing or $f' > 0$, hence one-to-one	2 pts
Part (b): Correct value $g(7) = 2$	2 pts
Part (c): Correct use of $g'(7) = 1/f'(2)$	2 pts
Part (c): Correct value $1/5$	1 pt
Part (d): Correct tangent line equation	2 pts

Question 100 Official Model Solution

★★★★★ VERY IMPORTANT

COMPREHENSIVE ANALYTICAL SOLUTION

(a) By the Chain Rule, $f'(x) = -\sin(x^2)\cdot 2x = \mathbf{-2x\sin(x^2)}$.

(b) Differentiate $f'(x) = -2x\cdot\sin(x^2)$ using the Product Rule and Chain Rule:
$f''(x) = -2\cdot\sin(x^2) + (-2x)\cdot\cos(x^2)\cdot 2x = \mathbf{-2\sin(x^2) - 4x^2\cos(x^2)}$.

(c) Horizontal tangent requires $f'(x) = 0$, that is $-2x\sin(x^2) = 0$.
This gives $x = 0$ or $\sin(x^2) = 0$. On $[0, \sqrt{\pi}]$, x^2 ranges over $[0, \pi]$, so $\sin(x^2) = 0$ when $x^2 = 0$ or $x^2 = \pi$.
Therefore **$x = 0$ and $x = \sqrt{\pi}$**.

AP SCORING CRITERIA & POINT ALLOCATION

Evaluation Criterion	Points
Part (a): Correct Chain Rule giving $-2x\sin(x^2)$	2 pts
Part (b): Correct Product Rule structure	2 pts
Part (b): Correct chain factor on $\cos(x^2)$ term	2 pts
Part (c): Sets $f'(x) = 0$ and solves $\sin(x^2) = 0$	2 pts
Part (c): Correct values $x = 0$ and $x = \sqrt{\pi}$	1 pt

Essential Unit 3 Rules

Core formulas required for mastery of composite, implicit, and inverse differentiation.

- **Chain Rule:** d/dx [f(g(x))] = f′(g(x))·g′(x).
- **Common chain forms:** d/dx [e^u] = e^u·u′ | d/dx [ln u] = u′/u | d/dx [u^n] = n·u^(n−1)·u′.
- **Trig chain forms:** d/dx [sin u] = cos u·u′ | d/dx [tan u] = $\sec^2 u \cdot u'$ | d/dx [sec u] = sec u tan u·u′.
- **Implicit Differentiation:** Differentiate both sides with respect to x, attaching dy/dx to every y-term, then solve for dy/dx.
- **Horizontal tangent:** numerator of dy/dx is 0. **Vertical tangent:** denominator of dy/dx is 0 (numerator nonzero).
- **Inverse Function Derivative:** $(f^{-1})'(b) = 1 / f'(f^{-1}(b))$. First find the input a with f(a) = b, then take 1/f′(a).
- **Inverse Trig:** $(\arcsin u)' = u'/\sqrt{1 - u^2}$ | $(\arccos u)' = -u'/\sqrt{1 - u^2}$ | $(\arctan u)' = u'/(1 + u^2)$.
- **Higher-Order:** f″ = derivative of f′. For motion, v(t) = s′(t) and a(t) = v′(t) = s″(t).
- **Speed test:** speed increases when v and a have the same sign; it decreases when their signs differ.

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